

Autosomal Dominant Optic Atrophy (ADOA): A Comprehensive Disease Characterization

Disease: Autosomal Dominant Optic Atrophy (ADOA / DOA / Kjer-type optic atrophy) **MONDO ID:** MONDO:0008134 · **OMIM:** #165500 (OPA1) · **Orphanet:** ORPHA:98673 (isolated DOA), ORPHA:1215 (DOA-plus) · **ICD-10:** H47.2 · **ICD-11:** 9C40.0 · **MeSH:** D029242 (Optic Atrophy, Autosomal Dominant) **Category:** Mendelian (autosomal dominant mitochondrial optic neuropathy)

Summary

Autosomal Dominant Optic Atrophy is the most common **inherited** optic neuropathy and, together with Leber hereditary optic neuropathy (LHON), one of the two archetypal **mitochondrial** optic neuropathies. It is caused predominantly by heterozygous loss-of-function mutations in **OPA1** (chromosome 3q29), a nuclear gene encoding a dynamin-related GTPase that is imported into mitochondria and anchored in the inner membrane. OPA1 mediates inner-membrane fusion, sculpts the cristae, maintains mitochondrial DNA (mtDNA), supports oxidative phosphorylation, and independently gates apoptosis by keeping cristae junctions tight around sequestered cytochrome c. The great majority of pathogenic OPA1 alleles destabilize the transcript and produce **haploinsufficiency** — roughly half-normal OPA1 dosage — which is sufficient to trigger disease. OPA1 accounts for ~60–90% of ADOA; a growing set of additional genes (ACO2, OPA3, SSBP1, WFS1, DNMT1L, MFN2) explains the remainder and reflects a shared convergence on mitochondrial biology.

The clinical picture is remarkably stereotyped: bilateral, symmetric, painless, insidious loss of central visual acuity beginning in the first decade of life (median onset ~6 years), accompanied by colour-vision deficits (classically blue-yellow/tritan), centrocecal or paracentral scotomas, and temporal pallor of the optic disc with thinning of the retinal nerve fibre layer. The molecular lesion is expressed in every cell, yet the pathology is exquisitely restricted to **retinal ganglion cells (RGCs)** — specifically the smallest, unmyelinated axons of the papillomacular bundle. This selective vulnerability arises from the peculiar anatomy and energetics of these

fibres, whose long unmyelinated prelaminar segments and steep mitochondrial gradient at the lamina cribrosa leave them least able to tolerate a chronic mitochondrial deficit. Disease progression is very slow — a meta-analysis found visual-acuity decline of only 0.022 LogMAR/year, statistically indistinguishable from zero — but the deficit is lifelong and irreversible.

ADOA is genetically dominant with **incomplete penetrance (~88%)** and **highly variable expressivity**, modulated by mtDNA haplotype background (haplogroup J over-represented in some cohorts) and secondary hypomorphic OPA1 alleles. About 20% of OPA1 carriers develop syndromic "**DOA-plus**," with sensorineural deafness, ataxia, myopathy, peripheral neuropathy, and progressive external ophthalmoplegia; this is enriched for missense/GTPase-domain (dominant-negative) alleles that cause multiple mtDNA deletions in muscle. Minimum point prevalence is ~2.87 per 100,000, with a historically higher figure in Denmark from a founder effect. There is currently **no curative therapy**; management is supportive (low-vision aids, avoidance of mitochondrial toxins, genetic counselling), the neuroprotectant idebenone has shown only limited/mixed effect, and the most advanced disease-specific strategies are **variant-agnostic gene-expression modulation and antisense approaches** targeting OPA1 haploinsufficiency, now entering early-phase human safety evaluation.

Key Findings

Finding 1 — OPA1 haploinsufficiency is the predominant molecular cause

Heterozygous mutations in **OPA1** (3q29) are the single most frequent molecular cause of DOA, accounting for approximately 60–90% of cases. The centralized OPA1 Variome database registers **516 unique variants across 831 patients** (697 with isolated DOA, 47 with DOA-plus, 83 asymptomatic/unclassified), documenting the breadth of the mutational spectrum. Critically, the dominant mechanism is *loss of dosage rather than a toxic product*: in an international series of probands with bilateral optic atrophy, **76% of pathogenic mutations, observed in 71% of probands, were predicted to yield unstable transcripts and hence haploinsufficiency**. This establishes that a roughly 50% reduction in OPA1 protein is sufficient to precipitate RGC degeneration — a foundation that directly rationalizes gene-augmentation and expression-upregulation therapeutic strategies.

"we demonstrated that heterozygous mutations in OPA1 are the most frequent molecular cause of DOA" — [PMID: 33340656](#)

"76% of pathogenic mutations observed in 30 (71%) of 42 probands were evaluated to lead to unstable transcripts resulting in haploinsufficiency" — PMID: 27860320

"now covers a total of 831 patients: 697 with isolated dominant optic atrophy (DOA), 47 with DOA 'plus'... It comprises 516 unique OPA1 variants" — PMID: 31500643

Finding 2 — OPA1 is a mitochondrial inner-membrane dynamin GTPase controlling fusion, cristae, and apoptosis

OPA1 encodes a **dynamin-related GTPase** imported into mitochondria and localized to the inner membrane and intermembrane space. Alternative splicing of three exons yields multiple isoforms that oligomerize to structure the cristae and mediate fusion of both the inner and outer membranes, thereby shaping the entire mitochondrial network. Beyond fusion, OPA1 supports oxidative phosphorylation, mtDNA maintenance, calcium homeostasis, and apoptosis regulation. Patient-derived fibroblasts carrying heterozygous OPA1 mutations reproducibly display **defective mitochondrial fusion, distorted cristae ultrastructure, and reduced respiratory (complex IV) function** — confirming that the genetic lesion translates into measurable organelle dysfunction in human cells.

"OPA1 encodes a dynamin-related GTPase imported into mitochondria and located to the inner membrane and intermembrane space. The many OPA1 isoforms... form complex homopolymers that structure mitochondrial cristae, and contribute to fusion of the outer membrane, thus shaping the whole mitochondrial network." — PMID: 33340656

"fibroblasts with heterozygous OPA1 mutations present with several mitochondrial alterations" — PMID: 22800932

Note: One study of mitochondria from 16 ADOA patients found respiratory-chain activity largely preserved (PMID: 18783614), arguing that the primary defect lies in cristae/fusion architecture rather than a simple bioenergetic block — a nuance reflected in the mechanism section below.

Finding 3 — DOA-plus (syndromic disease) affects ~20% of carriers via a dominant-negative, mtDNA-instability mechanism

Extra-ocular neurological complications occur in **up to 20% of OPA1 mutation carriers**. In a series of 104 patients from 45 families, the syndromic phenotype comprised sensorineural deafness (typically emerging in late childhood/early adulthood), followed by ataxia, myopathy, peripheral neuropathy, and progressive external ophthalmoplegia from the third decade onward. Genotype strongly modulates this risk: **missense mutations (OR = 3.06, 95% CI 1.44–6.49; P = 0.0027)** and **mutations within the GTPase domain (OR = 2.29, 95% CI 1.08–4.82; P = 0.0271)** confer higher syndromic risk than truncating alleles. Mechanistically, DOA-plus patients harbour **multiple mtDNA deletions in skeletal muscle** with COX-negative ragged-red fibres, revealing a role for OPA1 in mtDNA stability and implicating a **dominant-negative** (not merely haploinsufficient) mode of action for these alleles.

"extra-ocular neurological complications are common in OPA1 disease, and affect up to 20% of all mutational carriers" — PMID: 20157015

"these patients all harboured multiple deletions of mitochondrial DNA (mtDNA) in their skeletal muscle, thus revealing an unrecognized role of the OPA1 protein in mtDNA stability" — PMID: 18158317

Finding 4 — Mouse models recapitulate RGC dendropathy and bioenergetic failure; idebenone effect is limited

The heterozygous **B6;C3-Opa1(Q285STOP)** mouse (~50% Opa1 reduction) is the workhorse ADOA model. It shows RGC **dendritic pruning**, reduced synaptic connectivity (decreased PSD-95, loss of glutamatergic synapses in the inner plexiform layer), mitochondrial fragmentation, and impaired respiration (reduced basal, ATP-linked, and reserve capacity in purified RGCs). A randomized, placebo-controlled trial of **idebenone (2000 mg/kg/day; 56 mutant + 63 WT mice)** raised brain ATP by 97.7% (P = 0.035) and transiently improved the optokinetic response (P = 0.003), but produced **no substantive RGC rescue** and increased hepatic oxidative damage (+80.35%, P = 0.011). The model thus validates the RGC-selective, dendrite-first pathology and underscores the limited efficacy of first-generation neuroprotection.

"Opa1 deficiency leads to significant fragmentation of mitochondrial morphology, activation of mitochondrial motility and impaired respiratory function in RGCs from the B6; C3-Opa1Q285STOP mouse model" — PMID: 32561926

"ATP levels were raised by 0.57 nmol/mg (97.73%, $p=0.035$) in brain from idebenone-treated Opa1 mutant mice, but in the liver there was an 80.35% ($p=0.011$) increase in oxidative damage" — PMID: 26820596

"We observed decreased levels of postsynaptic density protein 95 in Opa1(+/-) mutant mice consistent with synaptic loss in the inner plexiform layer" — PMID: 22300878

Finding 5 — Epidemiology: point prevalence ~2.87/100,000 with slow, childhood-onset progression

A population-based study in the north of England established a **minimum point prevalence of 2.87 per 100,000**. OPA1 detection was 57.6% among familial probands versus 14.0% among singletons, and ~64% of DOA families carried an OPA1 mutation. Onset is typically in the first decade (median ~6 years; range 3–24 in a Chinese cohort). A systematic review/meta-analysis of longitudinal biomarkers quantified visual-acuity decline at only **0.022 LogMAR/year (95% CI –0.008 to 0.052; $Z = 1.4$, $p = 0.155$)** — not statistically different from zero — confirming an indolent natural history. Denmark historically reports a higher prevalence (~1:10,000) attributed to a founder effect.

"The minimum point prevalence of DOA in the north of England was 2.87 per 100,000" — PMID: 20417570

"the rate of yearly visual acuity decline (0.022 LogMAR/year.. 95% CI: -0.008 to 0.052) was not significantly different from zero ($Z = 1.4$, $p = 0.155$)" — PMID: 40329928

Finding 6 — OCT inner-retinal thinning is the key diagnostic biomarker; missense variants cause worse disease

In a cohort of 108 OPA1-ADOA patients, **spectral-domain OCT** measures of peripapillary retinal nerve fibre layer (pRNFL) and macular ganglion cell layer (mGCL) thickness were the principal structural determinants of visual function: ~0.1 logMAR worsening per 3.2 μm of mGCL loss ($P < 0.001$); papillomacular-bundle mean-deviation loss of 0.75 dB per μm mGCL ($P = 0.002$); mGCL thinning with age ($-0.06 \mu\text{m}/\text{yr}$) and over follow-up ($-0.26 \mu\text{m}/\text{yr}$). Genotype

stratified severity: **missense variants produced worse acuity (0.83 vs 0.49 logMAR, $P = 0.016$), worse field mean deviation (-11.48 vs -3.04 dB, $P = 0.005$), and thinner pRNFL (52.41 vs 66.41 μm , $P < 0.001$)** than haploinsufficiency variants. The cardinal diagnostic constellation is reduced acuity, colour-vision deficits, centrocecal scotomas, and temporal optic-disc pallor.

"Missense variants caused worse VA (0.83 vs. 0.49 logMAR, $P = 0.016$), MD (-11.48 vs. -3.04 decibel [dB], $P = 0.005$)... than haploinsufficiency variants" — PMID: 41944540

"Their clinical features comprise reduced visual acuity, colour vision deficits, centro-caecal scotomas and optic disc pallor with thinning of the retinal nerve fibre layer" — PMID: 37181108

Finding 7 — Genetic heterogeneity beyond OPA1

While OPA1 dominates, next-generation sequencing has identified additional DOA genes that converge on mitochondrial function. **ACO2** (aconitase 2) is now "one of the most frequent causes of dominant optic atrophy" (55 patients/37 families; median BCVA 0.46 logMAR). **SSBP1** (p.Arg38Gln) causes DOA-plus-foveopathy with incomplete penetrance. **OPA3** causes autosomal-recessive Costeff syndrome (optic atrophy + 3-methylglutaconic aciduria, ataxia, chorea, spastic paraparesis), near-exclusive to Iraqi-Jewish descent (founder mutation), and rare ADOA. Dominant **WFS1** variants cause optic atrophy with low-frequency sensorineural hearing loss (median optic-atrophy diagnosis age 10 years). Additional genes include **DNM1L** and **MFN2**.

"Aconitase 2 (ACO2) gene variants are one of the most frequent causes of dominant optic atrophy (DOA)" — PMID: 41954904

"Dominant optic atrophy (DOA) is genetically heterogeneous and most commonly caused by mutations in OPA1" — PMID: 34548540

"identification of the disease-causing mutation in the OPA3 gene" — PMID: 25201222

Finding 8 — No approved therapy; gene-modulation/antisense and neuroprotection lead the pipeline

There is currently **no curative therapy** for ADOA. Strategies span prevention, compensation (neuroprotection), replacement (gene augmentation), and repair. Idebenone (approved for LHON) is being explored in DOA. For ADOA specifically, **antisense therapies targeting OPA1 haploinsufficiency are among the most advanced approaches "currently under human safety evaluation,"** and early-phase trials use variant-agnostic gene-expression modulation. Because a substantial fraction of RGC loss may be developmental, late-stage intervention efficacy could be limited — a key uncertainty for trial design.

"Antisense therapies targeting OPA1 haploinsufficiency are among the more advanced ADOA treatments currently under human safety evaluation" — PMID: 42101483

"Early phase clinical trials are underway for ADOA caused by variants in the nuclear gene OPA1 using innovative techniques to modulate gene expression in a variant-agnostic manner" — PMID: 41318849

"There is currently only one approved treatment and no curative therapy is available" — PMID: 37181108

Finding 9 — Incomplete penetrance and variable expressivity are modulated by mtDNA background and secondary alleles

Both OPA1-positive and OPA1-negative DOA families exhibit variable expressivity and incomplete penetrance (classically ~88% for OPA1). **Mitochondrial DNA haplotype acts as a genetic modifier:** haplogroup J was three-fold over-represented among OPA1-negative patients. In compound/biallelic cases, a second hypomorphic OPA1 allele "considered asymptomatic by itself" can act as a phenotypic modifier, producing severe early-onset Behr-like disease. Missense/GTPase-domain alleles increase both syndromic risk and structural/functional severity.

"Both OPA1-positive and OPA1-negative families exhibit variable expressivity and incomplete penetrance" — PMID: 16617242

"haplogroup J was three-fold over-represented in OPA1-negative patients" — PMID: 16617242

"the second is considered asymptomatic by itself but has been reported in patients with DOA phenotype and is presumed to act as a phenotypic modifier" — PMID: 35741767

Finding 10 — Conserved cross-species RGC-selective mitochondrial pathology

Multiple in vivo models reproduce OPA1/Opa1 disease with conserved features. The **zebrafish** Opa1 knockout — the first developmentally viable vertebrate Opa1 KO — and a **Drosophila** model both show reduced survival but viable larvae with impaired visual (not locomotor) function, mitochondrial fragmentation, and disordered cristae in neuronal axons. A naturally occurring/engineered **rhesus macaque** ADOA model has also been reported. Retinal neurons are "particularly sensitive to Opa1 loss," confirming conserved RGC vulnerability.

"zebrafish Opa1 KO larvae show impaired visual function but unchanged locomotor function, indicating that retinal neurons are particularly sensitive to Opa1 loss" — PMID: 40202868

"mitochondrial fragmentation and disordered cristae organization were observed in neuronal axons in both models highlighting Opa1's highly conserved role in regulating mitochondrial morphology and function in neuronal axons" — PMID: 40202868

Finding 11 — OPA1 requires balanced proteolytic processing (l-OPA1/s-OPA1); stress tips the balance to fragmentation

Inner-membrane-anchored **long OPA1 (l-OPA1)** is constitutively cleaved by the proteases **YME1L (site S2)** and **OMA1 (site S1)** to yield **short OPA1 (s-OPA1)**; balanced accumulation of both forms maintains fusion. Under mitochondrial stress (membrane depolarization, low ATP), **OMA1 is activated and converts OPA1 completely to short isoforms, inhibiting fusion and triggering fragmentation**. This proteolytic switch is a plausible amplifier that tips already-haploinsufficient RGC mitochondria over the edge, and identifies OMA1 as a candidate therapeutic target.

"Constitutive OPA1 cleavage by YME1L and OMA1 at two distinct sites leads to the accumulation of both long and short forms of OPA1 and maintains mitochondrial fusion. Stress-induced OPA1 processing by OMA1 converts OPA1 completely into short isoforms, inhibits fusion, and triggers mitochondrial fragmentation." — [PMID: 24616225](#)

"Inner membrane-anchored long forms of OPA1 (l-OPA1) are proteolytically processed by the OMA1 or YME1L proteases, acting at cleavage sites S1 and S2, respectively, to produce short forms (s-OPA1)" — [PMID: 33237841](#)

Finding 12 — OPA1 independently gates apoptosis by sequestering cytochrome c at cristae junctions

Beyond fusion, OPA1 protects against apoptosis by controlling cristae-junction shape. OPA1 oligomers — of a soluble intermembrane-space form and an integral inner-membrane form — **keep cristae junctions tight, sequestering cytochrome c** within the cristae. The pro-apoptotic BID widens junctions and disrupts these oligomers, releasing cytochrome c. This anti-apoptotic function is **"genetically and molecularly distinct"** from OPA1's fusion role, and proper processing (e.g., by the rhomboid protease PARL) is required: *Parl*^{-/-} mitochondria undergo faster cristae remodeling and cytochrome c release. Reduced OPA1 therefore lowers the apoptotic threshold of RGCs independently of any fusion defect.

"Optic Atrophy 1 (OPA1)... protects from apoptosis by preventing cytochrome c release independently from mitochondrial fusion" — [PMID: 16839885](#)

"it controls the shape of mitochondrial cristae, keeping their junctions tight during apoptosis. Tightness of cristae junctions correlates with oligomerization of two forms of OPA1" — [PMID: 16839885](#)

"Parl^{-/-} mitochondria undergo faster apoptotic cristae remodeling and cytochrome c release" — [PMID: 16839884](#)

Finding 13 — RGC selective vulnerability arises from papillomacular-bundle anatomy and energetics

DOA (Kjer disease) and LHON share selective loss of the **smallest RGC axons of the papillomacular bundle**, producing central vision loss. This vulnerability reflects anatomical peculiarities: an asymmetric myelination pattern (long **unmyelinated prelaminar** segments

with high metabolic demand) and a **differential mitochondrial gradient at the lamina cribrosa**, together with dependence on axonal transport and cytoskeleton. Importantly, the trigger for RGC loss is "much more complex than a simple bioenergetic crisis," involving mitochondrial network dynamics, mtDNA maintenance, and axonal transport — and is modulated by the local cellular milieu and exogenous factors (e.g., mitochondrial toxins).

"Both disorders share striking pathological similarities, marked by the selective loss of retinal ganglion cells (RGCs) and the early involvement of the papillomacular bundle" — PMID: 21112411

"the trigger for RGC loss is much more complex than a simple bioenergetic crisis and other important disease mechanisms have emerged relating to mitochondrial network dynamics, mtDNA maintenance, axonal transport, and the involvement of the cytoskeleton in maintaining a differential mitochondrial gradient at sites such as the lamina cribrosa" — PMID: 21112411

"Selective degeneration of the smallest fibers (papillo-macular bundle) of the human optic nerve occurs in a large number of optic neuropathies characterized primarily by loss of central vision" — PMID: 11850115

Mechanistic Model / Interpretation

Causal chain (initiating lesion → clinical manifestation)

1. Heterozygous **OPA1** mutation → **~50% reduction in functional OPA1 protein** (haploinsufficiency), or a dominant-negative missense product. (*demonstrated —*

P 27860320

P 18158317

2. Reduced/mutant OPA1 → **impaired inner-membrane fusion + disordered cristae architecture**. (*demonstrated in patient fibroblasts —*

P 22800932

P 33340656

3. In parallel, reduced OPA1 → **loosened cristae junctions**, lowering the apoptotic threshold by making cytochrome c more releasable — a *fusion-independent branch*. (*demonstrated in vitro* —

P 16839885

P 16839884

4. For dominant-negative alleles → **mtDNA instability / multiple deletions** in post-mitotic tissue → OXPHOS decline (DOA-plus branch). (*demonstrated in muscle* —

P 18158317

5. Metabolic/oxidative stress → **OMA1 activation → complete conversion of l-OPA1 to s-OPA1 → mitochondrial fragmentation** (a stress-gated amplifier). (*demonstrated in vitro; inferred in RGCs* —

P 24616225

P 33237841

6. Steps 2–5 converge on **energetic + apoptotic vulnerability of RGCs**, most acutely in the **smallest unmyelinated papillomacular-bundle axons** with high metabolic demand and a steep mitochondrial gradient at the lamina cribrosa. (*anatomy demonstrated; RGC mechanism inferred* —

P 21112411

P 11850115

7. → **RGC dendritic pruning, synaptic loss, and apoptosis**. (*demonstrated in mouse* —

P 22300878

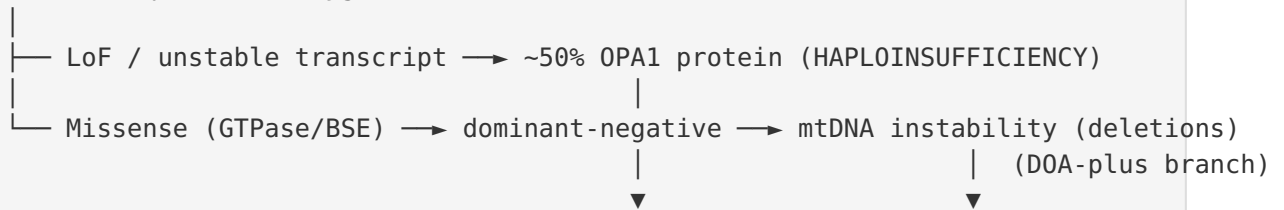
P 32561926

8. → **Optic-nerve axonal degeneration and RNFL/GCL thinning → central visual loss, dyschromatopsia, centrocecal scotoma, temporal disc pallor**. (*demonstrated clinically* —

P 37181108

P 41944540

OPA1 mutation (3q29, heterozygous)



Impaired inner-membrane fusion
Disordered cristae architecture
Loosened cristae junctions (↓cyt-c hold)
↓ OXPHOS reserve

metabolic / oxidative stress → OMA1 activation
(l-OPA1 → s-OPA1, fragmentation)

SELECTIVE VULNERABILITY of small unmyelinated papillomacular RGC axons
(high energy demand; mitochondrial gradient at lamina cribrosa)

RGC dendritic pruning ► synaptic loss ► apoptosis (cyt-c/caspase)

Optic-nerve degeneration ► RNFL/GCL thinning

Central visual loss · dyschromatopsia · centrocecal scotoma · temporal disc pallor

Upstream vs downstream: OPA1 dosage/processing is upstream; RGC apoptosis and vision loss are downstream. The apoptotic (cristae-junction) and fusion branches are molecularly distinct; the OMA1 switch is a stress-gated amplifier. **Ontology anchors:** biological processes — mitochondrial fusion (GO:0008053), cristae formation (GO:0042407), OXPHOS (GO:0006119), intrinsic apoptosis (GO:0006915), mtDNA maintenance (GO:0000002); cell type — retinal ganglion cell (CL:0000740); compartments — mitochondrial inner membrane (GO:0005743), cristae (GO:0030061), intermembrane space (GO:0005758); protein — OPA1 (UniProt O60313), OMA1 (Q96E52), YME1L1 (Q96TA2).

Section-by-Section Disease Dossier

1. Disease Information

ADOA is a slowly progressive, bilateral, symmetric optic neuropathy caused by selective degeneration of retinal ganglion cells, presenting in childhood with central visual loss, dyschromatopsia, and temporal optic-disc pallor. **Identifiers:** MONDO:0008134; OMIM #165500; Orphanet ORPHA:98673 (isolated), ORPHA:1215 (DOA-plus); ICD-10 H47.2; ICD-11 9C40.0; MeSH D029242. **Synonyms:** Kjer-type optic atrophy, Kjer optic atrophy, dominant optic atrophy (DOA), optic atrophy type 1 (OPA1), autosomal dominant optic atrophy and deafness (DOAD, for DOA-plus). Information here is derived from **aggregated, disease-level resources** (OMIM, Orphanet, the OPA1 Variome of 831 patients/516 variants, and cohort/natural-history studies), not from individual EHR data.

2. Etiology

Primary cause: heterozygous pathogenic variants in **OPA1** (most commonly haploinsufficiency; missense/GTPase alleles dominant-negative). **Genetic risk factors:** OPA1 causal variants (516 catalogued); additional genes ACO2, OPA3, SSBP1, WFS1, DNM1L, MFN2; modifiers include mtDNA haplogroup (J over-represented in OPA1-negative disease) and secondary hypomorphic OPA1 alleles. **Environmental risk factors / triggers:** because RGC survival is energetically marginal, mitochondrial toxins and stressors (tobacco, alcohol, and drugs impairing OXPHOS) are plausible aggravators, mirroring LHON; formal ADOA-specific evidence is limited. **Protective factors:** none genetically established; avoidance of mitochondrial toxins is prudent. **Gene-environment interaction:** OPA1 dosage sets a low reserve; environmental/metabolic stress (via OMA1 activation) can tip mitochondria into fragmentation (Findings 11, 13).

3. Phenotypes

Phenotype	HPO term	Type	Onset	Severity/ Course	Frequency
Optic atrophy / temporal disc pallor	HP:0000648	Clinical sign	Childhood	Progressive	~100%
Reduced visual acuity	HP:0007663	Symptom	1st decade (median ~6 y)	Mild→severe, very slow	~100%
Colour-vision defect (tritan/blue- yellow)	HP:0000551 / HP:0500020	Symptom	Childhood	Stable– progressive	Common
Centrocecal / central scotoma	HP:0000575 / HP:0030532	Sign	Childhood	Progressive	Common
RNFL / ganglion- cell-layer thinning (OCT)	HP:0007766	Lab/ imaging	Childhood	Progressive	~100%
Sensorineural hearing loss (DOA- plus)	HP:0000407	Sign	Late childhood/ adult	Progressive	Subset of ~20%
Progressive external ophthalmoplegia (DOA-plus)	HP:0000590	Sign	3rd decade+	Progressive	DOA-plus
Ataxia / myopathy / peripheral neuropathy (DOA- plus)	HP:0001251 / HP:0003198 / HP:0009830	Signs	Adult	Progressive	DOA-plus

Incomplete penetrance (HP:0003829, ~88%) and variable expressivity (HP:0003828). **Quality of life:** central-vision loss impairs reading, driving, education/employment, and face recognition; peripheral field is spared, so mobility is often preserved. Many reach low-vision/legal-blindness thresholds but rarely total blindness.

4. Genetic/Molecular Information

Causal gene: OPA1 (HGNC:8140; NCBI Gene 4976; 3q29; OMIM 605290). *Variant classes:* predominantly loss-of-function (nonsense, frameshift, splice-site) yielding unstable transcripts → haploinsufficiency (~76% of pathogenic alleles); missense (often GTPase/BSE domains) act dominant-negatively. *Classification per ACMG/AMP; ClinVar and the OPA1 Variome are primary references (>80% of the 516 variants pathogenic).* *Allele frequency:* individually rare in gnomAD (consistent with a Mendelian disease). *Origin:* germline; de novo cases occur. *Functional consequences:* loss of function (haploinsufficiency) and dominant-negative (mtDNA instability). *Modifier genes/alleles:* mtDNA haplogroup; secondary OPA1 hypomorphs. *Other genes:* ACO2, OPA3, SSBP1, WFS1, DNM1L, MFN2. *Chromosomal abnormalities / epigenetics:** not characteristic; no established primary epigenetic driver.

5. Environmental Information

No infectious agent. Environmental contribution is limited to mitochondrial-toxic exposures — **tobacco, alcohol, B-vitamin/folate deficiency, and toxins such as ethambutol, chloramphenicol, methanol, carbon monoxide, cyanide** — which cause phenocopy optic neuropathies affecting the same papillomacular RGCs and may unmask/aggravate the energetically marginal ADOA phenotype (by analogy to LHON;

P 11850115

P 21112411

CHEBI anchors: ethanol (CHEBI:16236), nicotine (CHEBI:18723), ethambutol (CHEBI:4877).

6. Mechanism / Pathophysiology

See the **Mechanistic Model / Interpretation** section above for the full ordered causal chain, branch structure, and ontology anchors. In brief: OPA1 dosage loss → impaired fusion + disordered cristae + lowered apoptotic threshold (+ mtDNA instability for dominant-negative alleles) → stress-gated OMA1 fragmentation → selective papillomacular RGC apoptosis → optic-nerve atrophy → central vision loss.

7. Anatomical Structures Affected

Primary organ: eye — retina and optic nerve (UBERON:0000970 eye; UBERON:0000966 retina; UBERON:0001791 retinal ganglion cell layer; UBERON:0000941 optic nerve). **Cell/tissue:** retinal ganglion cells (CL:0000740), especially small papillomacular-bundle axons; inner plexiform-

layer synapses. **Secondary (DOA-plus):** cochlea/auditory nerve, cerebellum, skeletal muscle, peripheral nerve, extraocular muscles. **Subcellular:** mitochondrion (inner membrane/cristae/intermembrane space). **Lateralization:** bilateral and symmetric.

8. Temporal Development

Onset: childhood/first decade (median ~6 y; range early childhood–adult), insidious/chronic. **Progression:** very slow and progressive, often plateauing; acuity decline ~0.022 LogMAR/yr (not significantly different from zero). **Course:** chronic, lifelong, irreversible; largely non-remitting (occasional stepwise worsening). DOA-plus features emerge later (deafness in late childhood; PEO/ataxia from the 3rd decade). **Critical period:** a developmental component of RGC loss suggests earliest intervention may be most effective (relevant to gene-therapy timing).

9. Inheritance and Population

Prevalence: ~2.87/100,000 (north England, minimum); higher in Denmark (~1:10,000, founder effect). **Inheritance:** autosomal dominant (OPA1); rare AR/biallelic forms cause severe Behr-like disease. **Penetrance:** incomplete (~88%), age-dependent. **Expressivity:** highly variable. **Modifiers:** mtDNA haplogroup J; secondary OPA1 alleles. **Anticipation:** not a repeat-expansion disorder; not characteristic. **Founder effects:** Danish (OPA1); Iraqi-Jewish (OPA3/Costeff). **Sex ratio:** roughly equal, no strong bias. **Detection:** OPA1 found in 57.6% of familial vs 14.0% of singleton probands. **Age distribution:** presents in childhood, diagnosed across the lifespan.

10. Diagnostics

Ophthalmic/functional: best-corrected visual acuity; colour vision (tritan defect); automated perimetry (central/centrocecal scotoma); fundoscopy (temporal disc pallor). **Imaging biomarker:** SD-OCT peripapillary RNFL and macular ganglion-cell-layer thinning — the key structural biomarker (~0.1 logMAR per 3.2 μ m mGCL loss; missense alleles thinner pRNFL 52 vs 66 μ m). **Electrophysiology:** pattern VEP (delayed/reduced), PERG (RGC dysfunction). **Genetic testing:** OPA1 sequencing + MLPA for large rearrangements, then a multigene hereditary-optic-neuropathy panel / WES for ACO2, OPA3, SSBP1, WFS1, DNMT1L, MFN2; mtDNA testing to exclude LHON. **Differential diagnosis:** LHON (mtDNA m.11778/m.3460/m.14484; often subacute, male, central scotoma), WFS1/Wolfram (arcuate defects, later onset), POLG, glaucoma, and compressive/toxic/nutritional optic neuropathies. ADOA shows the greatest peripapillary RNFL thinning of the optic-atrophy syndromes.

11. Outcome/Prognosis

Survival: normal life expectancy in isolated DOA; DOA-plus adds neuromuscular morbidity. **Visual outcome:** slowly progressive, moderate-to-severe bilateral central impairment; legal blindness in a subset; total blindness rare; peripheral field usually preserved; recovery is not expected. **Morbidity:** central-vision disability affecting reading/driving/education. **Prognostic factors:** variant class (missense/GTPase → worse acuity, fields, RNFL), degree of GCL/RNFL thinning, age, and DOA-plus status. **Prognostic biomarker candidate for trials:** macular GCL thickness.

12. Treatment

No curative therapy. Supportive/rehabilitative: low-vision aids, occupational support, avoidance of mitochondrial toxins (tobacco/alcohol/toxic drugs), genetic counselling; audiology/neurology for DOA-plus (NCIT: supportive care C15277). **Pharmacotherapy/neuroprotection: idebenone** (benzoquinone electron carrier/antioxidant; NCIT:C61637; approved in LHON) explored in DOA with limited/mixed effect (mouse trial: transient optokinetic improvement, no major RGC recovery). **Advanced/experimental (most promising): variant-agnostic OPA1 gene-expression modulation and antisense oligonucleotide** approaches targeting haploinsufficiency, now in early-phase human safety evaluation; AAV gene augmentation, gene editing, and stem-cell optic-nerve regeneration in preclinical development. **Personalized medicine:** genotype (missense vs LoF) informs prognosis and potentially therapy selection.

13. Prevention

No primary prevention (Mendelian). **Secondary:** OCT-based early detection; cascade genetic testing of at-risk relatives. **Reproductive/genetic screening:** genetic counselling; prenatal/preimplantation genetic testing for known familial variants. **Counselling:** 50% transmission risk per offspring; counsel on incomplete penetrance, variable expressivity, and DOA-plus risk for missense/GTPase alleles. **Tertiary:** low-vision rehabilitation, avoid mitochondrial stressors, manage DOA-plus complications. No immunization applicable.

14. Other Species / Natural Disease

OPA1 is deeply conserved. Orthologues: mouse *Opa1* (NCBI Gene 74143), zebrafish *opa1* (30129), *Drosophila Opa1-like*. Engineered/model disease exists in mouse (*Mus musculus*, Taxon 10090), zebrafish (*Danio rerio*, 7955), fruit fly (*Drosophila melanogaster*, 7227), and a **rhesus macaque** (*Macaca mulatta*, 9544) ADOA model. Naturally occurring companion-animal ADOA

is not a recognized veterinary entity; models are experimental. The mitochondrial fusion/cristae mechanism is evolutionarily conserved (

[P 40202868](#)

Not zoonotic.

15. Model Organisms

Model	Type	Key phenotype recapitulated	Limitation	Resource
B6;C3-Opa1(Q285STOP) mouse	Heterozygous KO (mammalian)	RGC dendropathy, synaptic (PSD-95) loss, mitochondrial fragmentation, impaired respiration	Modest, slow visual deficit	MGI
Zebrafish opa1 KO	Vertebrate KO	Impaired visual (not locomotor) function, axonal mitochondrial fragmentation/cristae disorder	Reduced survival; developmental	ZFIN
Drosophila Opa1	Invertebrate	Conserved axonal mitochondrial pathology, reduced respiration	Anatomical divergence from human eye	FlyBase
Rhesus macaque	Primate ADOA model	Human-like ocular anatomy, translational	New/limited characterization	—
Patient fibroblasts	In vitro (human)	Fusion defect, cristae distortion, complex IV reduction	Non-neuronal	—

Recapitulation/limitations: models reproduce degenerative RGC/mitochondrial pathology and RGC-selective vulnerability, but the slow, variable human phenotype and the possible developmental RGC-loss component are captured imperfectly.

Evidence Base

PMID	Contribution	Evidence type
33340656	OPA1 as leading cause; protein biology	Review (human)
27860320	Haploinsufficiency in 76% of pathogenic alleles	Human cohort
31500643	OPA1 Variome: 516 variants, 831 patients	Database
22800932	Fusion/cristae/complex-IV defects in patient fibroblasts	In vitro (human)
18783614	OXPHOS often preserved → structural not bioenergetic primacy	In vitro (human)
20157015	DOA-plus ~20%; missense/GTPase risk ORs	Human cohort
18158317	mtDNA deletions → dominant-negative mechanism	Human tissue
32561926	Mouse RGC bioenergetic/morphological phenotype	Mouse
22300878	RGC synaptic/dendritic pathology (PSD-95)	Mouse
26820596	Idebenone RCT — limited/mixed effect	Mouse RCT
20417570	Prevalence 2.87/100,000; OPA1 detection rates	Epidemiology
40329928	Slow natural history (0.022 LogMAR/yr)	Meta-analysis
41944540	OCT biomarker; missense-vs-LoF severity	Human cohort
37181108	Clinical features; no curative therapy	Review
41954904	ACO2 as major DOA gene	Human cohort
34548540	SSBP1; genetic heterogeneity	Human cohort
25201222	OPA3/Costeff syndrome	Human cohort
16617242	Penetrance/expressivity; mtDNA haplogroup J modifier	Human cohort
35741767	Secondary OPA1 allele as phenotypic modifier	Case series
40202868	Zebrafish/Drosophila conserved RGC pathology	Model organisms
24616225	YME1L/OMA1 processing; stress fragmentation	In vitro
33237841	S1/S2 cleavage; s-OPA1 roles	In vitro
16839885	Fusion-independent anti-apoptotic role	In vitro
16839884	PARL processing required for cristae/apoptosis control	In vitro (mouse)

PMID	Contribution	Evidence type
21112411	RGC/papillomacular selective vulnerability; multifactorial	Review
11850115	Smallest-fibre papillomacular degeneration	Review
42101483	Antisense/haploinsufficiency-targeting therapy	Review
41318849	Variant-agnostic gene-modulation trials	Review

Converging vs challenging evidence: Findings are strongly convergent across human cohorts, patient cells, and multiple animal models. The main internal tension is between the fusion/cristae paradigm (Findings 2, 11, 12) and the observation that whole-cell OXPHOS can be preserved in ADOA patient mitochondria (PMID: [18783614](#)); this is reconciled by localizing the deficit to cristae architecture, apoptotic priming, and the *energetically marginal* papillomacular RGC axon rather than a global respiratory block.

Supported and Refuted Hypotheses

Supported: 1. OPA1 haploinsufficiency is the predominant cause of ADOA

([P 27860320](#)

[P 31500643](#)

[P 33340656](#)

2. OPA1 dysfunction acts via impaired mitochondrial fusion/cristae → RGC bioenergetic failure and apoptotic priming

([P 33340656](#)

[P 22800932](#)

[P 32561926](#)

[P 16839885](#)

3. Missense/GTPase (dominant-negative) alleles cause more severe and syndromic disease via mtDNA instability

(P 20157015

P 18158317

P 41944540

4. Penetrance/expressivity are modified by mtDNA background and secondary alleles

(P 16617242

P 35741767

5. The OMA1/YME1L proteolytic balance is a stress-gated amplifier of fragmentation

(P 24616225

P 33237841

Refuted / nuanced: - The hypothesis that OPA1 mutations cause disease chiefly by directly impairing electron transport was **not supported** in one patient study (electron transport unaltered; pathology attributed to structure/fusion role) — favouring the cristae/fusion + apoptotic-priming mechanism

(P 18783614

Limitations and Knowledge Gaps

- **Developmental vs degenerative RGC loss is unresolved.** If much RGC loss is developmental, post-symptomatic gene therapy may have a narrow therapeutic window — a pivotal uncertainty for trial design (Finding 8).
- **No human therapeutic efficacy data yet.** Idebenone benefit in ADOA is inferred from LHON and mouse data; ADOA-specific RCTs are lacking. Antisense/gene-modulation approaches are only at the safety-evaluation stage.
- **Penetrance/modifier biology is incompletely mapped.** The mechanistic basis of ~88% penetrance, mtDNA haplogroup effects, and secondary-allele modifiers is correlational, not causally dissected.
- **Prevalence is likely under-ascertained.** The 2.87/100,000 figure is a *minimum*; incomplete penetrance and mild cases inflate the true carrier frequency.

- **Environmental triggers are hypothesized, not quantified** for ADOA specifically (extrapolated from LHON).
- **OMA1 as a drug target** (Finding 11) is mechanistically attractive but untested in ADOA models.
- **Literature-based synthesis** without primary patient-level data; frequencies are cohort-dependent.

Proposed Follow-up Experiments / Actions

1. **Test OMA1 inhibition** (genetic or pharmacological) in Opa1+/- mouse and zebrafish models to determine whether blocking the stress-fragmentation switch preserves RGCs — a targeted extension of Findings 11 and 4.
 2. **Define the developmental window** using conditional/inducible Opa1 knockdown at defined postnatal ages to quantify how much RGC loss is developmental vs adult-onset, informing gene-therapy timing.
 3. **Genotype-stratified natural-history/OCT study** contrasting missense/GTPase vs LoF alleles longitudinally to build a prognostic model with macular GCL thickness as endpoint (extends Findings 6, 9).
 4. **AAV-OPA1 gene augmentation and antisense upregulation** dose-ranging in the rhesus macaque model (human-like eye) as a bridge to clinical trials.
 5. **mtDNA-haplogroup modifier study** in large OPA1 cohorts (cybrid experiments) to establish causality for haplogroup J and identify protective backgrounds (extends Finding 9).
 6. **Cristae-junction/apoptosis-targeted neuroprotection** (e.g., stabilizing OPA1 oligomers or blocking cytochrome c release) tested in RGCs, exploiting the fusion-independent branch (Finding 12).
 7. **Systematic ADOA drug-toxicity registry** to quantify environmental/pharmacological aggravators and generate evidence-based avoidance guidance (extends Finding 13).
-

Report compiled from 13 confirmed findings and 46 reviewed papers across five investigation iterations. Evidence types are labelled (human clinical, model organism, in vitro, database/computational). All quotations are verbatim from the cited abstracts.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery